Conversational Turns Predict Later Language Better Than Raw Word Counts

In LENA recordings, adult–child conversational turns at 18–24 months predict school-age language and IQ after SES control more robustly than adult word counts; this is prediction, not proof that increasing turns causes those outcomes.

GRADE: MODERATE-PREDICTION; LOW-CAUSATION | STATUS: DRAFT | DOMAIN: LANGUAGE | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Language quantity and interactional quality are distinct. Automated daylong recordings can count adult words (AWC) and adult–child conversational turns (CTC). In a longitudinal LENA sample, turns at 18–24 months accounted for 14–27% of variance in IQ, verbal comprehension, and vocabulary about a decade later after socioeconomic status was controlled; adult word counts in the same window were weakened by that control.

EVIDENCE AND CAUSAL ASSESSMENT

Gilkerson et al. (2018), Pediatrics: 146 children with monthly daylong recordings; follow-up at 9–14 years (WISC-V, PPVT, EVT). Observational, automated measurement, SES covariate. Child-to-adult effects and shared family factors remain. This is stronger identification than a small transcribed SES contrast extrapolated to millions of words; it is not an RCT of "more turns."

EFFECT SIZE AND PERSISTENCE

Reported 14–27% variance at a decade is a prediction interval in one cohort, not a portable intervention effect. Persistence is the follow-up itself; causation of persistence is not shown.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: untested-in-GI (conversational turn-taking shares unmeasured genetic variance with parental verbal ability and child communicative drive; not yet tested in sibling-comparison or twin designs).

LENA classification error; who talks more may already be the more verbal child; SES residual confounding; LENA-using families are not a random population.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean families with fewer recorded words care less, or that a missed month of conversation writes a life. Hart & Risley's figure is not this card's claim.

WHAT ACTION IS REASONABLY JUSTIFIED

Support time and conditions for contingent exchange. Treating a word-count dashboard as parenting quality is not justified.

VERIFIED SOURCES & DOI IDENTIFIERS

- Gilkerson et al. (2018), Pediatrics (PMC6192025)